

TIME SERIES ANALYSIS & CLASSIFICATION

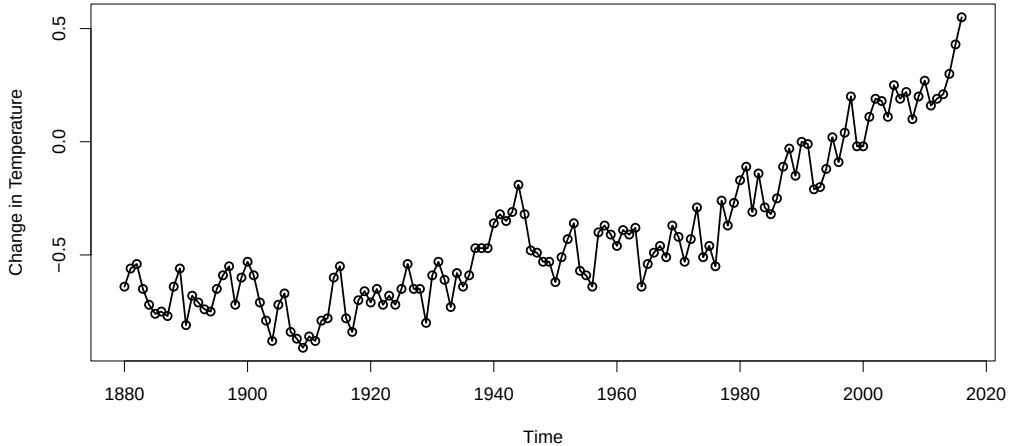
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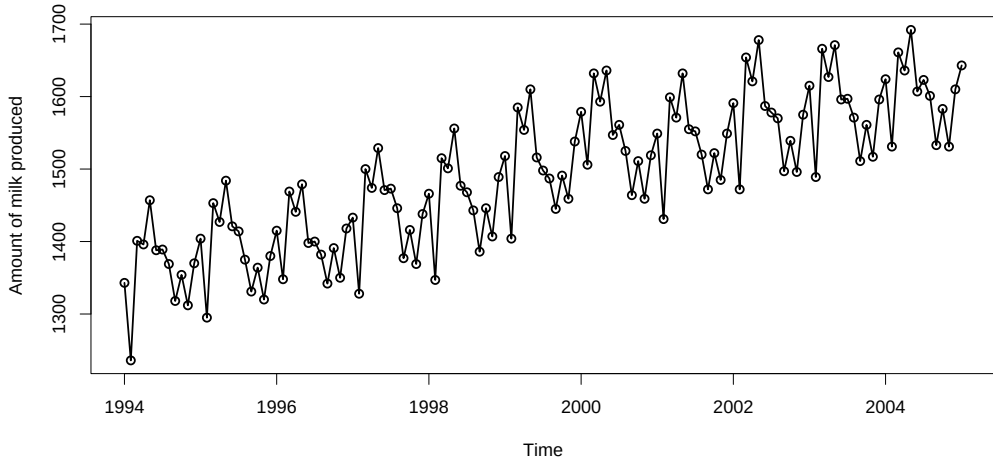
Chapter I: Introduction & Examples

1. Time series and stochastic processes
2. Means, Variances, and Covariances
3. Some Examples of Stochastic Processes
4. Stationarity

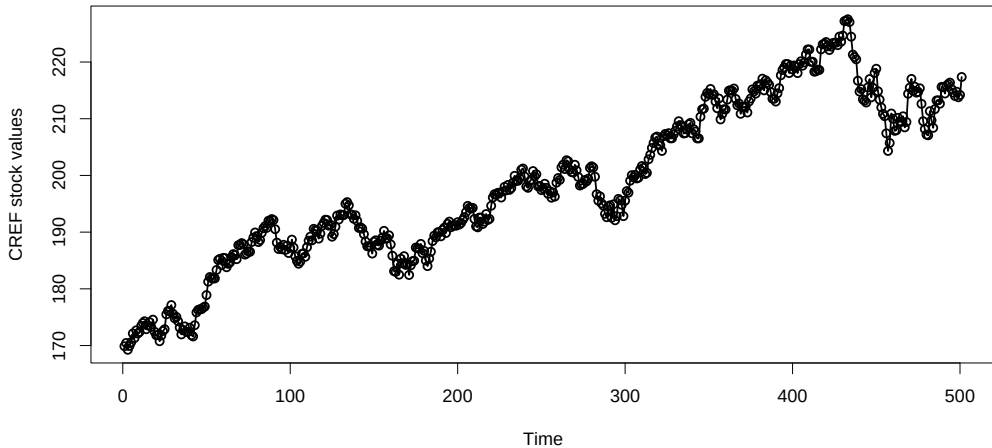
Examples: Global temperature data



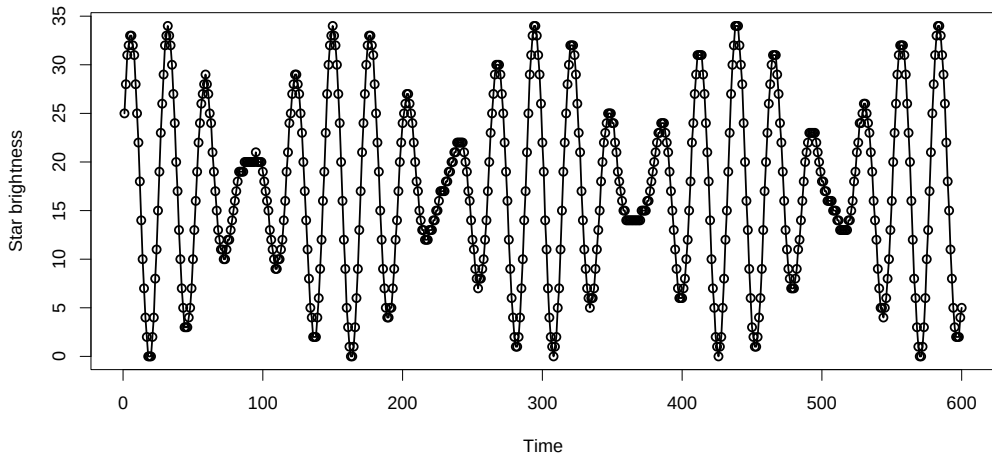
Examples: Milk production data



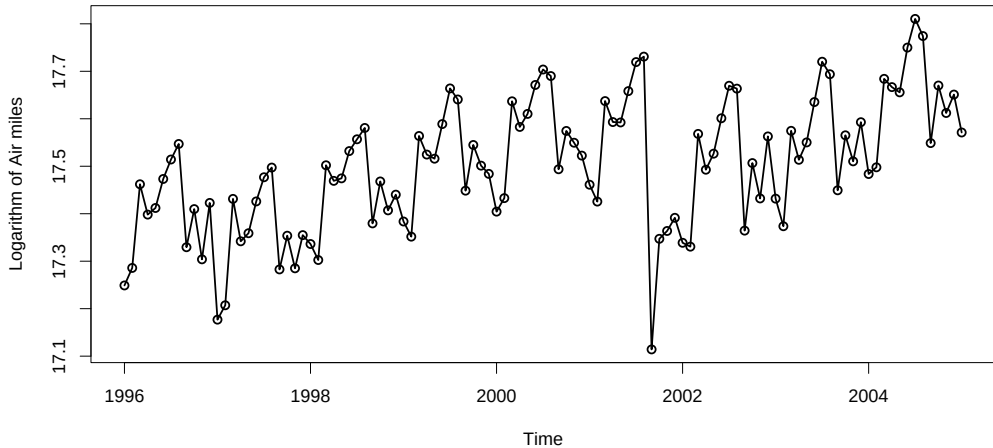
Examples: CREF stock data



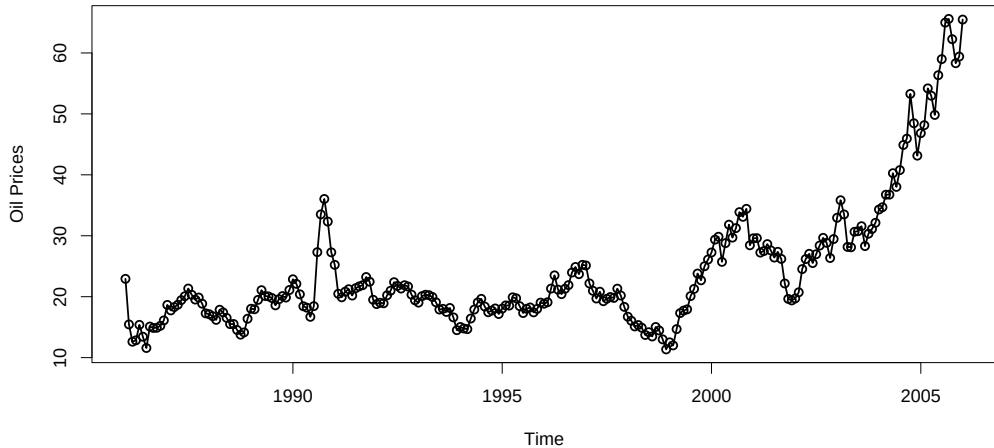
Examples: Star brightness data



Examples: Airline passenger mile data



Examples: Crude oil price data



Terminology

A time series is a sequence of ordered data. The ordering refers generally to time, but other orderings could be envisioned (e.g., over space, etc.). In this class, we will be concerned exclusively with time series that are:

- ▶ measured on a single continuous random variable Y ,
- ▶ equally spaced in discrete time; that is, we will have a single realization of Y at each second, hour, day, month, year, $\dots$ etc.

Ubiquity

Time series data arise in a variety of fields. Here are just a few examples:

- ▶ In business, we observe daily stock prices, weekly interest rates, quarterly sales, monthly supply figures, annual earnings, etc.
- ▶ In agriculture, we observe annual yields (e.g., crop production), daily crop prices, annual herd sizes, etc.
- ▶ In engineering, we observe electric signals, voltage measurements, etc.
- ▶ In natural sciences, we observe chemical yields, turbulence in ocean waves, earth tectonic plate positions, etc.

- ▶ In medicine, we observe ECG or EEG measurements on patients, drug concentrations, blood pressure readings, etc.
- ▶ In epidemiology, we observe the number of flu cases per day, the number of health-care clinic visits per week, annual tuberculosis counts, etc.
- ▶ In meteorology, we observe daily high temperatures, annual rainfall, hourly wind speeds, earthquake frequency, etc.
- ▶ In social sciences, we observe annual birth and death rates, accident frequencies, crime rates, school enrollments, etc.

Objectives

The purpose of time series analysis is twofold:

1. to model the stochastic (random) mechanism that gives rise to the series of data. To extract the behavior and features of the series,
2. to predict (forecast) the future values of the series based on the previous history.

Notes

- ▶ The analysis of time series data calls for a new way of thinking when compared to other statistical methods courses. Essentially, we get to see only a single measurement from a population (at time t) instead of a sample of measurements at a fixed point in time (cross-sectional data).

Notes

- ▶ The special feature of time series data is that they are not independent! Instead, observations are correlated through time.
 - Correlated data are generally more difficult to analyze.
 - Statistical theory in the absence of independence becomes markedly more difficult.
- ▶ Most classical statistical methods (e.g., regression, analysis of variance, etc.) assume that observations are statistically independent.

Notes

For example, in the simple linear regression model

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i,$$

we typically assume that the ϵ error terms are independent and identically distributed (iid) normal random variables with mean 0 and constant variance.

Notes

- ▶ There can be additional trends or seasonal variation patterns (seasonality) that may be difficult to identify and model.
- ▶ The data may be highly non-normal in appearance and be possibly contaminated by outliers

Modeling

Our goal in this part of the course is to build (and use) univariate time series models for data. This breaks down into different parts:

1. Model specification (identification):
 - Consider different classes of time series models for stationary processes.
 - Use descriptive statistics, graphical displays, subject matter knowledge, etc. to make sensible candidate selections.
 - Abide by the [Principle of Parsimony](#).

Modeling

2. Model fitting:

- Once a candidate model is chosen, estimate the parameters in the model.
- We will use least squares and/or maximum likelihood to do this.

3. Model diagnostics:

- Use statistical inference and graphical displays to check how well the model fits the data.
- This part of the analysis may suggest the candidate model is inadequate and may point to more appropriate models.

TIME SERIES AND STOCHASTIC PROCESSES

Terminology

The sequence of random variables $\{Y_t : t = 0, 1, 2, \dots\}$, or more simply denoted by $\{Y_t\}$, is called a stochastic process. It is a collection of random variables indexed by time t . That is,

Y_0 = value of the process at time $t = 0$

Y_1 = value of the process at time $t = 1$

Y_2 = value of the process at time $t = 2$

$\vdots$

Y_n = value of the process at time $t = n$.

The subscripts are important because they refer to which time period the value of Y is being measured.

Terminology

- ▶ A stochastic process can be described as a statistical phenomenon that evolves through time according to a set of probabilistic laws.
- ▶ A complete probabilistic time series model for $\{Y_t\}$ would specify all of the joint distributions of random vectors $\mathbf{Y} = \{Y_1, Y_2, \dots, Y_n\}$, for all $n = 1, 2, \dots$:

$P(Y_1 \leq y_1, Y_2 \leq y_2, \dots, Y_n \leq y_n)$, for all $y = (y_1, y_2, \dots, y_n)$ and $n = 1, 2, \dots$

Terminology

- ▶ This specification is not generally needed in practice. In this course, we specify only the first and second-order moments; i.e., expectations of the form $E(Y_t)$ and $E(Y_t Y_{t-k})$, for $k = 0, 1, 2, \dots$, and $t = 0, 1, 2, \dots$.
- ▶ Much of the important information in most time series processes is captured in these first and second moments (or, equivalently, in the means, variances, and covariances).

MEANS, VARIANCES, AND COVARIANCES

Terminology

- For the stochastic process $\{Y_t : t = 0, 1, 2, \dots\}$, the mean function is defined as

$$\mu_t = E(Y_t),$$

for $t = 0, 1, 2, \dots$. That is, μ_t is the theoretical (or population) mean for the series at time t .

Terminology

- ▶ The autocovariance function is defined as

$$\gamma_{t,s} = \text{cov}(Y_t, Y_s) = E(Y_t Y_s) - E(Y_t)E(Y_s),$$

for $t, s = 0, 1, 2, \dots$

- ▶ The autocorrelation function is given by

$$\rho_{t,s} = \text{corr}(Y_t, Y_s) = \frac{\text{cov}(Y_t, Y_s)}{\sqrt{\text{var}(Y_t) \text{var}(Y_s)}} = \frac{\gamma_{t,s}}{\sqrt{\gamma_{t,t} \gamma_{s,s}}}.$$

Terminology

- ▶ Values of $\rho_{t,s}$ near $\pm 1 \Rightarrow$ strong linear dependence between Y_t and Y_s .
- ▶ Values of $\rho_{t,s}$ near 0 $\Rightarrow$ weak linear dependence between Y_t and Y_s .
- ▶ Values of $\rho_{t,s} = 0 \Rightarrow Y_t$ and Y_s are **uncorrelated**.

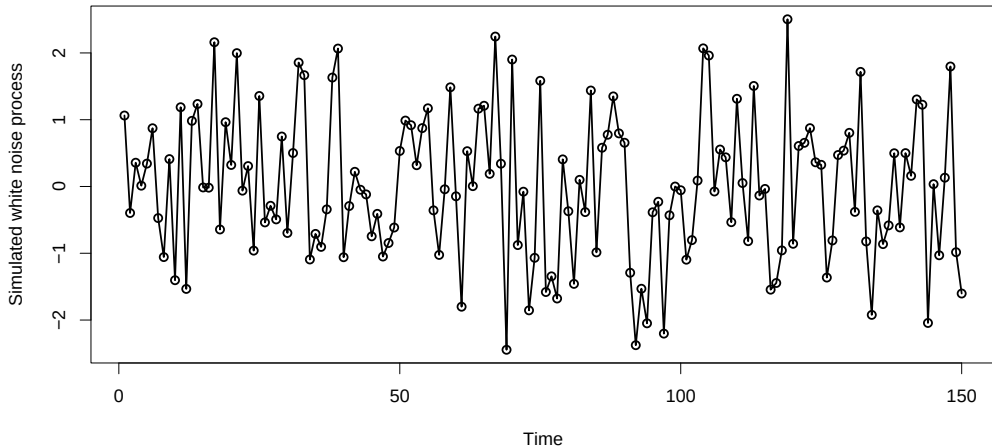
SOME EXAMPLES OF STOCHASTIC PROCESSES

Example 1: White noise

A stochastic process $\{e_t : t = 0, 1, 2, \dots\}$ is called a white noise process if it is a sequence of independent and identically distributed (iid) random variables with

$$\mathbb{E}(e_t) = \mu_e, \quad \text{Var}(e_t) = \sigma_e^2.$$

Example 1: plot



Example 1: Remarks

- ▶ Both μ_e and σ_e^2 are constant (free of t).
- ▶ It is often assumed that $\mu_e = 0$; that is, $\{e_t\}$ is a zero mean process.
- ▶ A slightly less restrictive definition would require that the e_t are uncorrelated (not independent).
- ▶ However, under normality; i.e., e_t are iid $\mathcal{N}(0, \sigma_e^2)$, this distinction becomes irrelevant (for linear time series models).

Example 1: Auto-covariance Function

- For $t = s$:

$$\text{Cov}(e_t, e_s) = \text{Cov}(e_t, e_t) = \text{Var}(e_t) = \sigma_e^2.$$

- For $t \neq s$:

$$\text{Cov}(e_t, e_s) = 0,$$

because the e_t 's are independent.

Thus, the autocovariance function of e_t is

$$\gamma_{t,s} = \begin{cases} \sigma_e^2, & \text{for } |t - s| = 0 \\ 0, & \text{for } |t - s| \neq 0 \end{cases}$$

Example 1: Autocorrelation Function

- For $t = s$:

$$\rho_{t,s} = \text{Corr}(e_t, e_s) = \text{Corr}(e_t, e_t) = \frac{\gamma_{t,t}}{\sqrt{\gamma_{t,t}\gamma_{t,t}}} = 1.$$

- for $t \neq s$:

$$\rho_{t,s} = (e_t, e_s) = \frac{\gamma_{t,s}}{\sqrt{\gamma_{t,t}\gamma_{s,s}}} = 0.$$

Thus, the autocorrelation function is:

$$\rho_{t,s} = \begin{cases} 1, & \text{for } |t - s| = 0 \\ 0, & \text{for } |t - s| \neq 0 \end{cases}$$

Important Remark 1

- ▶ A white noise process, by itself, is rather uninteresting for modeling real data. However, white noise processes still play a crucial role in the analysis of time series data!

Important Remark 2

Time series processes Y_t generally contain two different types of variation:

- ▶ systematic variation (that we would like to capture and model; e.g., trends, seasonal components, etc.)
- ▶ random variation (that is just inherent background noise in the process).

Important Remark 3

- ▶ Our goal as data analysts is to extract the systematic part of the variation in the data (and incorporate this into our model). If we do an adequate job of extracting the systematic part, then the only part left over should be random variation, which can be modeled as white noise.

Example 2: Random Walk

- Suppose that e_t is a zero mean white noise process with $\text{Var}(e_t) = \sigma_e^2$. Define:

$$Y_1 = e_1$$

$$Y_2 = e_1 + e_2$$

$$\vdots$$

$$Y_n = e_1 + e_2 + \dots + e_n.$$

Example 2: Random Walk

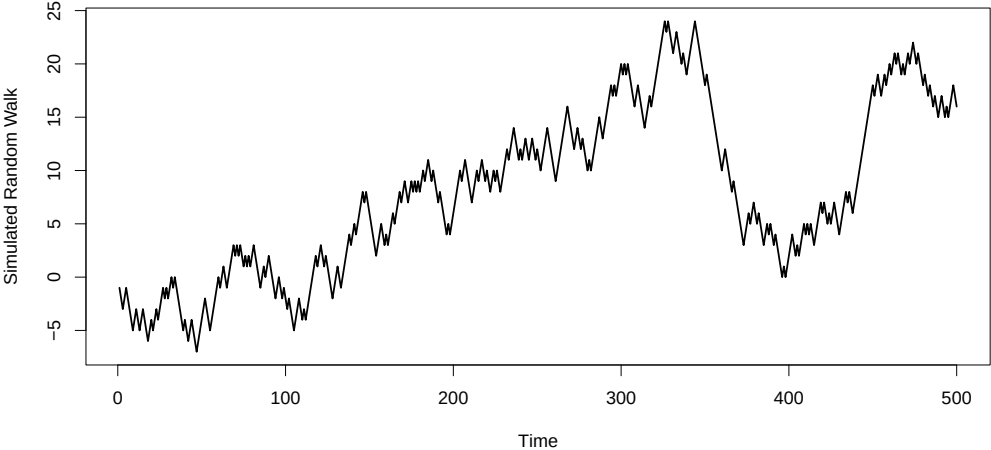
- ▶ By this definition, note that we can write, for $t > 1$,

$$Y_t = Y_{t-1} + e_t,$$

where $E(e_t) = 0$ and $\text{var}(e_t) = \sigma_e^2$.

- ▶ The process $\{Y_t\}$ is called a **random walk** process.
- ▶ Random walk processes are used to model stock prices, movements of molecules in gases and liquids, animal locations, etc.

Example 2: Plot



Example 2: Mean Function

- The mean of Y_t is:

$$\begin{aligned}\mu_t &= \mathbf{E}(Y_t) \\ &= \mathbf{E}(e_1 + e_2 + \dots + e_t) \\ &= \mathbf{E}(e_1) + \mathbf{E}(e_2) + \dots + \mathbf{E}(e_t) = 0.\end{aligned}$$

That is, Y_t is a zero mean process.

Example 2: Variance Function

- The variance of Y_t is:

$$\begin{aligned}\text{Var}(Y_t) &= \text{Var}(e_1 + e_2 + \dots + e_t) \\ &= \text{Var}(e_1) + \text{Var}(e_2) + \dots + \text{Var}(e_t) = t\sigma_e^2,\end{aligned}$$

because $\text{Var}(e_1) = \text{Var}(e_2) = \dots = \text{Var}(e_t) = \sigma_e^2$, and $\text{Cov}(e_t, e_s) = 0$ for all $t \neq s$.

Example 2: Autocovariance Function

- For $t \leq s$, the autocovariance of Y_t and Y_s is:

$$\begin{aligned}\gamma_{t,s} &= \text{Cov}(Y_t, Y_s) \\ &= \text{Cov}(e_1 + \dots + e_t, e_1 + \dots + e_t + e_{t+1} + \dots + e_s) \\ &= \text{Cov}(e_1 + e_2 + \dots + e_t, e_1 + e_2 + \dots + e_t) \\ &\quad + \text{Cov}(e_1 + e_2 + \dots + e_t, e_{t+1} + \dots + e_s) \\ &= \sum_{i=1}^t \text{Cov}(e_i, e_i) + \sum \sum_{1 \leq i \neq j \leq t} \text{Cov}(e_i, e_j) \\ &= \sum_{i=1}^t \text{Var}(e_i) = \sigma_e^2 + \sigma_e^2 + \dots + \sigma_e^2 = t\sigma_e^2.\end{aligned}$$

Because $\gamma_{t,s} = \gamma_{s,t}$, the autocovariance function for a random walk process is

$$\gamma_{t,s} = t\sigma_e^2, \text{ for } 1 \leq t \leq s.$$

Example 2: Autocorrelation Function

- ▶ For $1 \leq t \leq s$, the autocorrelation function for a random walk process is

$$\rho_{t,s} = (Y_t, Y_s) = \frac{\gamma_{t,s}}{\sqrt{\gamma_{t,t}\gamma_{s,s}}} = \frac{t\sigma_e^2}{\sqrt{t\sigma_e^2 s\sigma_e^2}} = \sqrt{\frac{t}{s}}.$$

Note that:

- ▶ when t is closer to s , the autocorrelation $\rho_{t,s}$ is closer to 1. That is, two observations Y_t and Y_s close together in time are likely to be close together, especially when t and s are both large (later on in the series).
- ▶ On the other hand, when t is far away from s (that is, for two points Y_t and Y_s far apart in time), the autocorrelation is closer to 0.

Example 3

- Suppose that $\{e_t\}$ is a zero mean white noise process with $\text{var}(e_t) = a_e^2$. Define:

$$Y_t = \frac{1}{3}(e_t + e_{t-1} + e_{t-2}),$$

that is, Y_t is a running (or moving) average of the white noise process (averaged across the most recent 3 time periods).

Example 3: Mean Function

- The mean of Y_t is:

$$\begin{aligned}\mu_t = \mathbb{E}(Y_t) &= \mathbb{E}\left[\frac{1}{3}(e_t + e_{t-1} + e_{t-2})\right] \\ &= \frac{1}{3}[\mathbb{E}(e_t) + \mathbb{E}(e_{t-1}) + \mathbb{E}(e_{t-2})] = 0,\end{aligned}$$

because $\{e_t\}$ is a zero-mean process. $\{Y_t\}$ is a zero mean process.

Example 3: Variance Function

- The variance of Y_t is

$$\begin{aligned}\text{Var}(Y_t) &= \text{Var}\left[\frac{1}{3}(e_t + e_{t-1} + e_{t-2})\right] \\&= \frac{1}{9} \text{Var}(e_t + e_{t-1} + e_{t-2}) \\&= \frac{1}{9} [\text{Var}(e_t) + \text{Var}(e_{t-1}) + \text{Var}(e_{t-2})] \\&= \frac{1}{9} (3 \text{Var}(e_t)) = \frac{1}{3} \sigma_e^2,\end{aligned}$$

because $\text{Var}(e_t) = \sigma_e^2$ for all t and because e_t , e_{t-1} , and e_{t-2} are uncorrelated (all covariance terms are zero).

Example 3: Autocovariance Function

We need to consider different cases:

- ▶ Case 1: If $s = t$, then:

$$\gamma_{t,s} = \gamma_{t,t} = \text{Cov}(Y_t, Y_t) = \text{Var}(Y_t) = \frac{1}{3} \sigma_e^2.$$

Example 3: Autocovariance Function

- Case 2: If $s = t + 1$, then

$$\begin{aligned}\gamma_{t,s} &= \gamma_{t,t+1} = \text{Cov}(Y_t, Y_{t+1}) \\ &= \text{Cov} \left[\frac{1}{3}(e_t + e_{t-1} + e_{t-2}), \frac{1}{3}(e_{t+1} + e_t + e_{t-1}) \right] \\ &= \frac{1}{9} [\text{Cov}(e_t, e_t) + \text{Cov}(e_{t-1}, e_{t-1})] \\ &= \frac{1}{9} [\text{Var}(e_t) + \text{Var}(e_{t-1})] = \frac{2}{9} \sigma_e^2.\end{aligned}$$

Example 3: Autocovariance Function

- Case 3: If $s = t + 2$, then

$$\begin{aligned}\gamma_{t,s} &= \gamma_{t,t+2} = (Y_t, Y_{t+2}) \\ &= \text{Cov} \left[\frac{1}{3}(e_t + e_{t-1} + e_{t-2}), \frac{1}{3}(e_{t+2} + e_{t+1} + e_t) \right] \\ &= \frac{1}{9} \text{Cov}(e_t, e_t) \\ &= \frac{1}{9} \text{Var}(e_t) = \frac{1}{9} \sigma_e^2.\end{aligned}$$

Example 3: Autocovariance Function

- ▶ Case 4: If $s > t + 2$, then $\gamma_{t,s} = 0$ because Y_t and Y_s will have no common white noise error terms.
- ▶ Because $\gamma_{t,s} = \gamma_{s,t}$, the autocovariance function can be written as:

$$\gamma_{t,s} = \begin{cases} \sigma_e^2/3, & |t - s| = 0 \\ 2\sigma_e^2/9, & |t - s| = 1 \\ \sigma_e^2/9, & |t - s| = 2 \\ 0, & |t - s| > 2. \end{cases}$$

Example 3: Autocorrelation Function

- Recall that the autocorrelation function is :

$$\rho_{t,s} = (Y_t, Y_s) = \frac{\gamma_{t,s}}{\gamma_{t,t}\gamma_{s,s}}.$$

- Because $\gamma_{t,t} = \gamma_{s,s} = \frac{\sigma_e^2}{3}$, the autocorrelation function for this process is :

$$\rho_{t,s} = \begin{cases} 1, & |t-s|=0 \\ 2/3, & |t-s|=1 \\ 1/3, & |t-s|=2 \\ 0, & |t-s|>2. \end{cases}$$

Notes

- ▶ Observations Y_t and Y_s that are 1 unit apart in time have the same autocorrelation regardless of the values of t and s .
- ▶ Observations Y_t and Y_s that are 2 units apart in time have the same autocorrelation regardless of the values of t and s .
- ▶ Observations Y_t and Y_s that are more than 2 units apart in time are uncorrelated

STATIONARITY

Stationarity

NOTE: Stationarity is a very important concept in the analysis of time series data.

- ▶ A time series is said to be stationary if there is no systematic change in mean (no trend), if there is no systematic change in variance, and if strictly periodic variations have been removed.
- ▶ In other words, the properties of one section of the data are much like those of any other section.

Stationarity

IMPORTANCE: Much of the theory of time series is concerned with stationary time series. For this reason, time series analysis often requires one to transform a nonstationary time series into a stationary one to use this theory.

- For example, it may be of interest to remove the trend and seasonal variation from a set of data and then try to model the variation in the residuals (the pieces left-over after this removal) by means of a stationary stochastic process.

Stationarity

- ▶ The stochastic process $\{Y_t : t = 0, 1, 2, \dots, n\}$ is said to be **strictly stationary** if the joint distribution of

$$Y_{t_1}, Y_{t_2}, \dots, Y_{t_n}$$

is the same as

$$Y_{t_1-k}, Y_{t_2-k}, \dots, Y_{t_n-k}$$

for all time points $t_1, t_2, \dots, t_n$ and for all time lags k .

- ▶ In other words, shifting the time origin by an amount k has no effect on the joint distributions, which must therefore depend only on the intervals between $t_1, t_2, \dots, t_n$. This is a very strong condition.

Stationarity: Implication

Since the above condition holds for all sets of time points $t_1, t_2, \dots, t_n$, it must hold when $n = 1$; i.e., there is only one time point.

- ▶ This implies Y_t and Y_{t-k} have the **same marginal distribution** for all t and k .
- ▶ Because these marginal distributions are the same, we have for all t and k :

$$E(Y_t) = E(Y_{t-k})$$

$$\text{Var}(Y_t) = \text{Var}(Y_{t-k}),$$

- ▶ Therefore, for a strictly stationary process, both $\mu_t = E(Y_t)$ and $\gamma_{t,t} = \text{Var}(Y_t)$ are **constant** over time.

Stationarity: Implication

Since the above condition holds for all sets of time points $t_1, t_2, \dots, t_n$, it must hold when $n = 2$; i.e., there are only two time points.

- ▶ This implies (Y_t, Y_s) and (Y_{t-k}, Y_{s-k}) have the same joint distribution for all t, s , and k .
- ▶ Because these joint distributions are the same,

$$\text{Cov}(Y_t, Y_s) = \text{Cov}(Y_{t-k}, Y_{s-k}),$$

for all t, s , and k .

Stationarity: Implication

- Therefore, for a strictly stationary process, for $k = s$,

$$\gamma_{t,s} = \text{Cov}(Y_t, Y_s) = \text{Cov}(Y_{t-s}, Y_0) = \text{Cov}(Y_0, Y_{t-s}).$$

But, also, for $k = t$, we have

$$\text{Cov}(Y_t, Y_s) = \text{Cov}(Y_0, Y_{s-t}).$$

Putting the last two results together, we have

$$\gamma_{t,s} = \text{Cov}(Y_t, Y_s) = \text{Cov}(Y_0, Y_{|t-s|}) = \gamma_{0,|t-s|}.$$

This means that the covariance between Y_t and Y_s does not depend on the actual values of t and s ; it only depends on the time difference $|t - s|$.

New Notation

- ▶ For a (strictly) stationary process, the covariance $\gamma_{t,s}$ depends only on the time difference $|t - s|$. The quantity $|t - s|$ is the distance between time points Y_t and Y_s .
- ▶ In other words, the covariance between Y_t and any observation $k = |t - s|$ time points from it only depends on the **lag k** .
Therefore, we write:

$$\gamma_k = \text{Cov}(Y_t, Y_{t-k}),$$

$$\rho_k = (Y_t, Y_{t-k}).$$

New Notation

- ▶ We use this simpler notation only when we refer to a process which is stationary. Note that by taking $k = 0$, we have

$$\gamma_0 = \text{Cov}(Y_t, Y_t) = \text{Var}(Y_t),$$

Also,

$$\rho_k = \text{Corr}(Y_t, Y_{t-k}) = \frac{\gamma_k}{\gamma_0}.$$

Summary

For a process which is (strictly) stationary,

1. The mean function $\mu_t = E(Y_t) = \mu$ is constant throughout time; i.e., μ_t is free of t .
2. The covariance between any two observations depends only on the time lag between them; i.e., $\gamma_{t,t-k} = \gamma_k$ depends only on k (not on t).

Remark

- ▶ Strict stationarity is a condition that is much too restrictive for most applications. Moreover, it is difficult to assess the validity of this assumption in practice. Rather than impose conditions on all possible (marginal and joint) distributions of a process, we will use a milder form of stationarity that only deals with the first two moments.

Definition

The stochastic process Y_t : $t = 0, 1, 2, \dots, n$ is said to be **weakly stationary** (or second-order stationary) if :

1. The mean function $\mu_t = E(Y_t) = \mu$ is **constant** throughout time; i.e., μ_t is free of t .
2. The covariance between any two observations **depends only on the time lag between them**; i.e., $\gamma_{t,t-k} = \gamma_k$ depends only on k (not on t).

Nothing is assumed about the collection of joint distributions of the process. Instead, we only are specifying the characteristics of the first two moments of the process.

Realisation

- ▶ Clearly, strict stationarity implies weak stationarity. It is also clear that the converse to statement is not true, in general.
- ▶ However, if we append the additional assumption of multivariate normality (for the Y_t process), then the two definitions do coincide; that is,
weak stationarity + multivariate normality $\Rightarrow$ strict stationarity.

Convention

- ▶ For the purpose of modeling time series data in this course, we will rarely (if ever) make the distinction between strict stationarity and weak stationarity. When we use the term stationary process, this is understood to mean that the process is weakly stationary.

Example 1

We now reexamine the time series models introduced in the last section. Suppose that $\{e_t\}$ is a white noise process. That is, $\{e_t\}$ consists of iid random variables with $E(e_t) = \mu_e$ and $\text{Var}(e_t) = \sigma_e^2$, both constant (free of t). In addition, the autocovariance function $\gamma_k = \text{Cov}(Y_t, Y_{t-k})$ is given by

$$\gamma_k = \begin{cases} \sigma_e^2, & \text{for } k = 0 \\ 0, & \text{for } k \neq 0 \end{cases}$$

which is free of time t (i.e., γ_k depends only on k). Thus, a white noise process is stationary.

Example 2

Suppose that $\{Y_t\}$ is a random walk process. That is,

$$Y_t = Y_{t-1} + e_t,$$

where $\{e_t\}$ is white noise with $E(e_t) = 0$ and $\text{Var}(e_t) = \sigma_e^2$. We calculated $\mu_t = E(Y_t) = 0$, for all t (which is free of t). However,

$$\text{Cov}(Y_t, Y_{t-k}) = \text{Cov}(Y_{t-k}, Y_t) = (t-k)\sigma_e^2,$$

which clearly depends on time t . Thus, a random walk process is **not stationary**.

Example 3

Suppose that $\{Y_t\}$ is a moving average process given by

$$Y_t = \frac{1}{3}(e_t + e_{t-1} + e_{t-2}),$$

where $\{e_t\}$ is zero mean white noise with $\text{Var}(e_t) = \sigma_e^2$. We calculated $\mu_t = E(Y_t) = 0$ (which is free of t), and

$$\gamma_k = \begin{cases} \sigma_e^2/3, & k = 0 \\ 2\sigma_e^2/9, & k = 1 \\ \sigma_e^2/9, & k = 2 \\ 0, & k > 2. \end{cases}$$

Because $\text{Cov}(Y_t, Y_{t-k})$ is free of time t , this moving average process is stationary.